Resolvent norm bound for the Langevin dynamics generator under Poincare assumptions on the invariant measure and momentum measure

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Editorial note. Journal numbering: the counted theorem prints as Theorem 3.3 in the publisher PDF, the third theorem-family item of Section 3 under the shared family counter (Assumption 3.1, Assumption 3.2 precede it at source lines 581 and 593); the card seeds the section counter with 3 and the shared family counter cdrthm with 2, so the counted theorem prints as 3.3 exactly as in the journal.

Editorial note. Equation renumbering: the card-local numbered equations are the statement's eq:bounds_invL_Langevin and the proof's eq:A^*A_Langevin, eq:op_for_Ass5, eq:action_cLa2, eq:Sij_Tij, which print as (3.1)-(3.5) in the delivery baseline (journal (3.6)-(3.10) of Section 3). The context equation eq:regularization condition follows the counted proof and prints as (3.6) in the baseline (journal (3.5)); all cross-references, including the proof's references to eq:Pi_1 (journal (2.5), stubbed) and to the regularization condition, resolve through the two engine passes. The baseline renumbering is declared here for the record.

Editorial note. Citations: the counted proof cites DMS15 (Proposition 5) and ST18 (Lemma A.4) at source line 698; the publisher bibliography is shipped whole (bibliography_backend none, thebibliography file copied into the task directory) and both keys resolve.

Editorial note. External references: hypotheses ass:Poincare_nu (3.1), ass:Poincare_kappa (3.2), ass:A_inv (2.2), ass:macro_coercivity (2.3) and ass:technical_ass (2.6) are stubbed as framework hypotheses whose verification is the counted proof's work; the input theorem thm:bounds_Linv (2.7) is stubbed as the framework result the proof applies; the projector definition eq:Pi_1 (2.5) and the forward pointer to Proposition 3.5 are stubbed for reference resolution. None of these is counted by this card or by a sibling card.

Editorial note. Preamble composition: three ranges – the editorial numbering adapter first (family re-declaration on the cdrthm anchor counter, English names, numberwithin), then the retained preamble's provenance comments (lines 1-6) and its body (lines 8-99). Line 7, the active cedram documentclass, is excluded because the wrapper supplies the class; lines 100-160 (title, alttitle, authors, keywords, abstracts and front matter) are excluded as publisher class-metadata commands that carry no mathematical content. hyperref is declared through extra_packages; the wrapper's automatic amsmath, amssymb,

amsthm and graphicx load precedes the preamble, and the retained amssymb usepackage line is an inert duplicate. All notation macros used by the delivered spans are defined in the retained preamble itself; no mathematical macro, symbol, statement or proof step was added, changed or removed.

Editorial note. Occupancy and claim evidence: claimkey_index carries no ahl.129 key; the doi_public_occupancy gate sweep for doi 10.5802/ahl.129 ran PASS with occupied=0 in the 2026-10-15 morning gate over the full 30-candidate queue, and the census v3.12 row for this unit reads pairing own with occupied false.

Allowed standard prerequisites (editorial).

- The standing Langevin setting of Section 3.1 and the Section 2 block notation: D the position space with potential V, beta the inverse temperature, nu the Gibbs measure e^{-beta V} dq / Z on D, kappa the momentum measure e^{-beta U} dp / Z_kappa, mu the product measure, the Hamiltonian and fluctuation/dissipation generators cLham (journal (3.2)) and cLFD (journal (3.3)), the projector Pi_0 (journal (3.4)), the block decomposition cLa_subplus0 = Pi_subplus cLa Pi_0 with the inverse (cLa_subplus0^* cLa_subplus0)^{-1} of Assumption 2.2, the projectors Pi_1, Pi_2 and the generator split cL = cLs + cLa with cLs = gamma cLFD, cLa = cLham. the Section 3.1 opening prose (source lines 545-579) and the Section 2 block notation, outside the counted spans. Fixes D, beta, V, U, nu, kappa, mu and every block operator, projector and constant appearing in the counted statement and proof.

Source reference. Original reference 3.1: sources/ahl-129-hypocoercivity-with-schur-complements-2022/full-original-native.tex, source lines 581–591. This stub points at Assumption 3.1 of the journal (source lines 581-591), a hypothesis of the framework rather than a result proved in the source; the stub carries the journal number for reference resolution and is not counted by this card or by a sibling card.

Source reference. Original reference 3.2: sources/ahl-129-hypocoercivity-with-schur-complements-2022/full-original-native.tex, source lines 593–599. This stub points at Assumption 3.2 of the journal (source lines 593-599), a hypothesis of the framework; its moment bounds input the boundedness step at the close of the counted proof. Not counted by this card or by a sibling card.

Source reference. Original reference 2.7: sources/ahl-129-hypocoercivity-with-schur-complements-2022/full-original-native.tex, source lines 490–495. This stub points at the abstract framework theorem of Section 2 (source lines 490-495), which the counted proof applies at its opening and close; it is stated and proved in the source outside this card and is not counted here.

Source reference. Original reference 2.2: sources/ahl-129-hypocoercivity-with-schur-complements-2022/full-original-native.tex, source lines 382–384. This stub points at the invertibility hypothesis of Section 2 (source lines 382-384), a hypothesis verified, not proved, by the counted proof; not counted by this card or by a sibling card.

Source reference. Original reference 2.3: sources/ahl-129-hypocoercivity-with-schur-complements-2022/full-original-native.tex, source lines 392–397. This stub

points at the macroscopic coercivity hypothesis of Section 2 (source lines 392-397), a hypothesis whose verification is part of the counted proof's work; the assumption statement itself is not counted by this card.

Source reference. Original reference 2.6: sources/ahl-129-hypocoercivity-with-schur-complements-2022/full-original-native.tex, source lines 484–486. This stub points at the technical boundedness hypothesis of Section 2 (source lines 484-486), a hypothesis whose verification is the core of the counted proof; the assumption statement itself is not counted by this card.

Source reference. Original equation 2.5: sources/ahl-129-hypocoercivity-with-schur-complements-2022/full-original-native.tex, source lines 404–406. This stub resolves the proof's reference 'in view of (2.5) and (3.2)' (source line 664); it is a definition of the Section 2 block projector, not a result, and is not counted by this card.

Source reference. Original reference 3.5: sources/ahl-129-hypocoercivity-with-schur-complements-2022/full-original-native.tex, source lines 736–750. This stub resolves the proof's forward pointer 'see also the proof of Proposition ... below' (source line 698); the proposition is counted by the sibling card S9-ahl129-estimate and is not counted here.

Theorem 3.3. *Suppose that Assumptions 3.1 and 3.2 hold true. Then, the resolvent of the generator of the Langevin dynamics satisfies the following bound:*

$$\begin{aligned}\left\|\mathcal{L}^{-1}\right\| &\leqslant \frac{2\beta\gamma}{\lambda_{\min}(\mathcal{M})K_\nu^2}\left\|\Pi_1\mathcal{L}_{\mathrm{FD}}\Pi_1\right\| \\ &+\frac{4\beta}{\gamma K_\kappa^2}\left(\frac{3}{4}+\left\|\Pi_+\mathcal{L}_{\mathrm{ham}}^2\Pi_0\left(\mathcal{A}_{+0}^*\mathcal{A}_{+0}\right)^{-1}\right\|^2+\gamma^2\left\|\Pi_2\mathcal{L}_{\mathrm{FD}}\Pi_1\mathcal{L}_{\mathrm{ham}}\Pi_0\left(\mathcal{A}_{+0}^*\mathcal{A}_{+0}\right)^{-1}\right\|^2\right),\end{aligned} \tag{3.1}$$

where $\lambda_{\min}(\mathcal{M}) > 0$ denotes the smallest eigenvalue of the symmetric positive definite matrix

$$\mathcal{M} = \int_{\mathbb{R}^d} \nabla^2 U \, d\kappa = \beta \int_{\mathbb{R}^d} \nabla U \otimes \nabla U \, d\kappa.$$

Proof. We apply Theorem 2.7, and check to this end that Assumptions 2.2, 2.3 and 2.6 hold true. The Poincaré inequality in the momentum variable implies that Assumption 2.2 holds with

$$s = \gamma \frac{K_\kappa^2}{\beta}.$$

To check Assumptions 2.3 and 2.6, we use $\partial_{p_i}^* = -\partial_{p_i} + \beta\partial_{p_i}U$ and $\partial_{q_i}^* = -\partial_{q_i} + \beta\partial_{q_i}V$ to rewrite the generators as

$$\mathcal{S} = \gamma\mathcal{L}_{\mathrm{FD}} = -\frac{\gamma}{\beta}\sum_{i=1}^d \partial_{p_i}^*\partial_{p_i}, \qquad \mathcal{A} = \mathcal{L}_{\mathrm{ham}} = \frac{1}{\beta}\sum_{i=1}^d \partial_{p_i}^*\partial_{q_i} - \partial_{q_i}^*\partial_{p_i}.$$

We also use the following rules

$$\left[\partial_{p_i}, \partial_{q_j}\right] = 0, \qquad \partial_{p_i}\Pi_0 = 0, \qquad \Pi_0\partial_{p_i}^* = 0, \qquad \left[\partial_{p_i}, \partial_{p_j}^*\right] = \beta\partial_{p_i,\,p_j}^2 U,$$

to obtain

$$\begin{aligned}\mathcal{A}_{+0}^*\mathcal{A}_{+0} &= \frac{1}{\beta^2}\sum_{i=1}^d\sum_{j=1}^d \Pi_0\partial_{q_i}^*\partial_{p_i}\partial_{p_j}^*\partial_{q_j}\Pi_0 \\ &= \frac{1}{\beta}\Pi_0\nabla_q^*\mathcal{M}\nabla_q\Pi_0 \\ &= \frac{1}{\beta}\sum_{i,j=1}^d \mathcal{M}_{ij}\Pi_0\partial_{q_i}^*\partial_{q_j}\Pi_0.\end{aligned} \tag{3.2}$$

To prove that $\mathcal{M}$ is a symmetric positive definite matrix, we write, for $\xi \in \mathbb{R}^d$,

$$\xi^T\mathcal{M}\xi = \beta\int_{\mathbb{R}^d}\left|\xi^T\nabla U\right|^2 d\kappa.$$

If the latter quantity was 0 for some element $\xi \in \mathbb{R}^d\backslash\{0\}$, then ∇U would be 0 in the direction of ξ, and hence U would be constant in this direction, which would contradict the assumption $\mathrm{e}^{-\beta U} \in L^1(\mathbb{R}^d)$. We then obtain, for a smooth function φ with compact support and average 0 with respect to μ,

$$\begin{aligned}\left\|\mathcal{A}_{+0}\varphi\right\|_{L^2(\mu)}^2 &= \frac{1}{\beta}\int_{\mathcal{D}}\left(\nabla_q\Pi_0\varphi\right)^T\mathcal{M}\nabla_q\Pi_0\varphi\, d\nu \\ &\geqslant \frac{\lambda_{\min}(\mathcal{M})}{\beta}\left\|\nabla_q\Pi_0\varphi\right\|_{L^2(\nu)}^2 \geqslant \frac{\lambda_{\min}(\mathcal{M})K_\nu^2}{\beta}\left\|\Pi_0\varphi\right\|_{L^2(\mu)}^2,\end{aligned}$$

where we used the Poincaré inequality for ν in the last step. This shows that Assumption 2.3 holds true with

$$a = K_\nu\sqrt{\frac{\lambda_{\min}(\mathcal{M})}{\beta}}.$$

We finally turn to Assumption 2.6. Note first that

$$-\mathcal{A}_{+0}^*\mathcal{L}_{\mathrm{FD}}\mathcal{A}_{+0} = \frac{1}{\beta}\nabla_q^*\mathcal{N}\nabla_q\Pi_0,$$

with

$$\mathcal{N}_{ij} = \int_{\mathbb{R}^d}\left[\left(\nabla^2 U\right)^2\right]_{ij} d\kappa = \sum_{k=1}^d\int_{\mathbb{R}^d}\left(\partial^2_{p_i,\,p_k}U\right)\left(\partial^2_{p_k,\,p_j}U\right)\, d\kappa,$$

so that, in view of (2.5) and (3.2),

$$-\Pi_1\mathcal{L}_{\mathrm{FD}}\Pi_1 = \frac{1}{\beta}\sum_{i,j=1}^d T_{ij}\widetilde{T}_{ij},$$

with

$$T_{ij} = \partial_{p_i}^*\Pi_0\partial_{p_j}, \qquad \widetilde{T}_{ij} = \partial_{q_i}\left(\nabla_q^*\mathcal{M}\nabla_q\right)^{-1}\nabla_q^*\mathcal{N}\nabla_q\left(\nabla_q^*\mathcal{M}\nabla_q\right)^{-1}\partial_{q_j}^*,$$

where $\widetilde{T}_{ij}$ is seen as an operator on $L^2(\nu)$. Note that T_{ij} is bounded since it is the composition of operators of the form $\Pi_0 \partial_{p_k}$ and their adjoints, such operators being bounded because, by an integration by parts,

$$\Pi_0 \partial_{p_k} \varphi = \beta \int_{\mathbb{R}^d} \varphi \, \partial_{p_k} U \, d\kappa,$$

so that

$$\left\| \Pi_0 \partial_{p_k} \varphi \right\|_{L^2(\mu)} \leqslant \beta \left\| \partial_{p_k} U \right\|_{L^2(\mu)} \left\| \varphi \right\|_{L^2(\mu)}.$$

The operators $\widetilde{T}_{ij}$ are also easily seen to be bounded, as a linear combination of composition of operators of the form $(\nabla_q^* \mathcal{M} \nabla_q)^{-1/2} \partial_{q_j}^*$ (which are bounded from $L^2(\nu)$ to $L^2_0(\nu)$, the subspace of functions of functions of $L^2(\nu)$ with average 0 with respect to ν) and their adjoints. This allows to conclude that $\mathcal{S}_{11} = \gamma \Pi_1 \mathcal{L}_{\mathrm{FD}} \Pi_1$ is bounded.

Let us next consider

$$\mathcal{L}_{21} \mathcal{A}_{10} \big(\mathcal{A}_{+0}^* \mathcal{A}_{+0} \big)^{-1} = \Pi_2 \mathcal{S} \mathcal{A} \Pi_0 \big(\mathcal{A}_{+0}^* \mathcal{A}_{+0} \big)^{-1} + \Pi_2 \mathcal{A}^2 \Pi_0 \big(\mathcal{A}_{+0}^* \mathcal{A}_{+0} \big)^{-1}. \tag{3.3}$$

We start by the second operator on the right hand side of the latter equality. Since $\Pi_2 = \Pi_2 \Pi_+$, it suffices in fact to prove that $\Pi_+ \mathcal{A}^2 \Pi_0 \left(\mathcal{A}_{+0}^* \mathcal{A}_{+0} \right)^{-1}$ is bounded. First, for a smooth function φ with compact support, we compute $\mathcal{A}^2 \Pi_0 \varphi = \nabla U^T (\nabla_q^2 \Pi_0 \varphi) \nabla U - \nabla V^T (\nabla^2 U) \nabla_q \Pi_0 \varphi$, and therefore

$$\Pi_+ \mathcal{A}^2 \Pi_0 \varphi = \sum_{i,j=1}^d \mathcal{U}_{ij} \partial^2_{q_i, \, q_j} \Pi_0 \varphi - \mathcal{V}_{ij} \left(\partial_{q_i} V \right) \partial_{q_j} \Pi_0, \tag{3.4}$$

with

$$\mathcal{U}_{ij} = \left[\left(\partial_{p_i} U \right) \left(\partial_{p_j} U \right) - \frac{1}{\beta} \mathcal{M}_{ij} \right], \qquad \mathcal{V}_{ij} = \partial^2_{p_i, \, p_j} U - \mathcal{M}_{ij}. \tag{3.5}$$

The boundedness of the operator $\Pi_+ \mathcal{A}^2 \Pi_0 \left(\mathcal{A}_{+0}^* \mathcal{A}_{+0} \right)^{-1}$ then follows from the fact that the operators $\Pi_+ \partial^2_{q_i, \, q_j} \Pi_0 \left(\mathcal{A}_{+0}^* \mathcal{A}_{+0} \right)^{-1}$ and $\Pi_+ \left(\partial_{q_i} V \right) \partial_{q_j} \Pi_0 \left(\mathcal{A}_{+0}^* \mathcal{A}_{+0} \right)^{-1}$ are bounded (in view of (3.6), see respectively [DMS15, Proposition 5] and [ST18, Lemma A.4]; see also the proof of Proposition 3.5 below).

Let us finally bound the first operator on the right hand side of (3.3), using the following equality as operators from $\mathcal{H}_0$ to $\mathcal{H}_2$:

$$\Pi_2 \mathcal{S} \mathcal{A} \Pi_0 \big(\mathcal{A}_{+0}^* \mathcal{A}_{+0} \big)^{-1} = - \frac{\gamma}{\beta^2} \sum_{i=1}^d \Pi_2 \mathscr{T}_i \partial_{q_i} \big(\mathcal{A}_{+0}^* \mathcal{A}_{+0} \big)^{-1},$$

$$\mathscr{T}_i = \sum_{j=1}^d \partial^*_{p_j} \partial_{p_j} \partial^*_{p_i} \Pi_0.$$

The result then follows from the fact that $\partial_{q_i} (\mathcal{A}_{+0}^* \mathcal{A}_{+0})^{-1}$ is bounded from $\mathcal{H}_0$ to $\mathcal{H}$, and

$$\mathscr{T}_i = \beta \sum_{j=1}^d \left[\beta \left(\partial_{p_j} U \right) \left(\partial^2_{p_i, \, p_j} U \right) - \partial^3_{p_i, p_j, \, p_j} U \right] \Pi_0$$

is a bounded operator on $L^2(\mu)$, because of the $L^2(\kappa)$-bounds on the derivatives of U (see Assumption 3.2). The estimate (3.1) finally follows by Theorem 2.7. □

$$\Delta V \leqslant c_1 d + \frac{c_2 \beta}{2} |\nabla V|^2 , \qquad \left|\nabla^2 V\right|^2 := \sum_{i,j=1}^{d} \left|\partial_{x_i} \partial_{x_j} V\right|^2 \leqslant c_3^2 \left(d + |\nabla V|^2\right), \tag{3.6}$$

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